American Computer Science League

2021-2022 ● Contest 2: Fibonacci Cypher ● Junior Division

PROBLEM: Use this Fibonacci cypher to encode a given message.

The Fibonacci sequence starts with two non-negative integers and every term after that is the sum of the two previous numbers. Therefore, for this problem, our Fibonacci sequence will be 1, 2, 3, 5, 8, 13, … so that all terms are unique numbers. A Fibonacci cypher uses these numbers and an inputted 'key' (a lower-case letter in the alphabet) to encode a message into a series of numbers. Another person can decipher the message by knowing the 'key' that was used. For each character in the message, integers are generated by using the following steps:

1. Find the Fibonacci number with the same index as the character (e.g. the 1st character in the inputted string uses the Fibonacci number 1 and the 5th character uses the number 8).
2. By using the inputted key, find a new letter that is offset by the Fibonacci number found in Step 1. Wrap around to the beginning of the alphabet if necessary (e.g. if the key = 'p' and the Fibonacci number is 21, then the new letter is 'k').
3. Find the encoded number for each letter in the message by adding the ASCII code of the character in the string and the ASCII code of the letter found in Step 2.

EXAMPLE: For the string "ACSL c2" and the letter 'h' the numbers are found as shown:

Fibonacci offset	Offset ASCII Code	String ASCII code	Calculation
h + 1 = i	i = 105	A = 65	65 + 105 = 170
h + 2 = j	j = 106	C = 67	67 + 106 = 173
h + 3 = k	k = 107	S = 83	83 + 107 = 190
h + 5 = m	m = 109	L = 76	76 + 109 = 185
h + 8 = p	p = 112	space = 32	32 + 112 = 144
h + 13 = u	u = 117	c = 99	99 + 117 = 216
h + 21 = c	c = 99	2 = 50	50 + 99 = 149

The encoded message is "170 173 190 185 144 216 149".

INPUT: There are 5 sets of data. Each set contains 2 data values: 1) a lower-case alphabetic character for the 'key' and 2) a string for the actual message to be encoded. We guarantee that the message will be no longer than 30 characters. Any typeable character on the keyboard except for double quotes may be included in the message.

OUTPUT: For each line of data, print the integers, each separated by a single space, that represent the encoded message.

American Computer Science League

2021-2022 ● Contest 2: Fibonacci Cypher ● Junior Division

SAMPLE INPUT:

```
h ACSL c2
s Python Programming
a Fibonacci Numbers
j Help ME!
z It is 9:30 in the morning.
```

SAMPLE OUTPUT:

1. 170 173 190 185 144 216 149
2. 196 238 234 224 208 212 142 177 232 211 206 228 199 210 223 205 209 213
3. 168 204 198 213 215 207 217 204 205 140 189 239 219 207 223 222 222
4. 179 209 217 223 146 196 170 147
5. 170 214 131 206 219 141 174 162 150 155 142 226 219 140 237 211 207 149 210 233 215 211 211 221 224 156

American Computer Science League

2021-2022 ● Contest 2: Fibonacci Cypher ● Junior Division

TEST DATA

TEST INPUT:

```
h ACSL Contest 2
b Madam, I'm Adam!
a Java programs are the best.
d What are we really doing?
g This is the Fibonacci Cypher!
```

TEST OUTPUT:

1. 170 173 190 185 144 184 210 222 223 216 233 219 149 166
2. 176 197 201 200 215 155 151 179 140 218 144 162 211 207 206 142
3. 172 196 218 199 137 222 232 216 203 222 208 231 225 141 219 222 208 150 218 201 203 134 205 213 237 227 156
4. 188 206 200 221 140 210 235 209 135 230 215 131 227 213 196 219 218 242 137 200 216 210 220 218 162
5. 188 209 211 223 143 221 213 143 222 218 218 134 186 220 200 225 223 195 207 202 213 140 180 239 214 221 217 218 150

American Computer Science League

2021-2022 ● Contest 2: Fibonacci Cypher ● Junior Division

题目: 使用**斐波那契**加密方法**给以下**特定的**信息编码**加密。

斐波那契数列从**两个非负整数开始，后边每一项的值都是前两项的和。本题中所用到的斐波那契数列为** 1,2, 3, 5, 8, 13, …，其中每一项**都是不同的数字。斐波那契**加密方法**使用这些数字和**一个**输入“ 密钥 ”（字母表中的一个小写字母）将一**条**信息编码成一连串的数字。**其他人可通过已使用**的“ 密钥 ”**来**破译信息。信息中的每个字符都可以根据以下步骤来生成**对应的**整数：**

1. 找出**与该字符具有相同**索引**的斐波那契数（如输入字符串中的第一个字符用斐波那契数** 1 来表示**，第五个字符用** 8 来**表示）。**
2. **根据输入**的**密钥，找出**对步**骤** 1 中**斐波那契数**进行偏移操作的新字母，如有**必要，可以**重新绕到**字母表**的**开头**处（比如**密钥是“** p **”，斐波那契数是** 21，**那么新的字母即为“** k **”）。**
3. 通过将字符串中的字符的 ASCII 码与步骤 2 中找到的字母的 ASCII 码相加，找出信息中每个字母的编码数字。

示例: 对于字符串 “ACSL c2” **和字母**‘h’ **， 可以通过以下方式**获得对应的编码数字。

斐波那契偏移	偏移ASCII 码	字符串ASCII 码	计算结果
h + 1 = i	i = 105	A = 65	65 + 105 = 170
h + 2 = j	j = 106	C = 67	67 + 106 = 173
h + 3 = k	k = 107	S = 83	83 + 107 = 190
h + 5 = m	m = 109	L = 76	76 + 109 = 185
h + 8 = p	p = 112	space = 32	32 + 112 = 144
h + 13 = u	u = 117	c = 99	99 + 117 = 216
h + 21 = c	c = 99	2 = 50	50 + 99 = 149

编码后**的信息**是 “170 173 190 185 144 216 149”。

输入: 共有 5 组**数据，每组包含** 2 个**数据值：** 1) 一个**小写字母用作“ 密钥 ”；** 2) 一条**需要编码的字符串**信息。我们保证**每**条**信息的长度不会超过** 30 个**字符。除双引号外，键盘上的任意字符都可能在信息中出现。**

输出: 每行数据都输出整数，每个整数之**间由空格隔开，表示编码**后**的信息。**

American Computer Science League

2021-2022 ● Contest 2: Fibonacci Cypher ● Junior Division

样本输入：

```
h ACSL c2
s Python Programming
a Fibonacci Numbers
j Help ME!
z It is 9:30 in the morning.
```

预期输出：

1. 170 173 190 185 144 216 149
2. 196 238 234 224 208 212 142 177 232 211 206 228 199 210 223 205 209 213
3. 168 204 198 213 215 207 217 204 205 140 189 239 219 207 223 222 222
4. 179 209 217 223 146 196 170 147
5. 170 214 131 206 219 141 174 162 150 155 142 226 219 140 237 211 207 149 210 233 215 211 211 221 224 156

American Computer Science League

2021-2022 ● Contest 2: Fibonacci Cypher ● Junior Division

测试数据

测试输入：

```
h ACSL Contest 2
b Madam, I'm Adam!
a Java programs are the best.
d What are we really doing?
g This is the Fibonacci Cypher!
```

测试输出：

1. 170 173 190 185 144 184 210 222 223 216 233 219 149 166
2. 176 197 201 200 215 155 151 179 140 218 144 162 211 207 206 142
3. 172 196 218 199 137 222 232 216 203 222 208 231 225 141 219 222 208 150 218 201 203 134 205 213 237 227 156
4. 188 206 200 221 140 210 235 209 135 230 215 131 227 213 196 219 218 242 137 200 216 210 220 218 162
5. 188 209 211 223 143 221 213 143 222 218 218 134 186 220 200 225 223 195 207 202 213 140 180 239 214 221 217 218 150